

Closed-Form Evaluation of $\sin 15^\circ$ and $\cos 15^\circ$

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Solution

Let

$$A = 15^\circ,$$

and write

$$s = \sin A, \quad c = \cos A.$$

Since

$$6 \times 15^\circ = 90^\circ,$$

we have

$$\cos 6A = \cos 90^\circ = 0.$$

Using the multiple-angle formula for $\cos 6A$,

$$\cos 6A = 32c^6 - 48c^4 + 18c^2 - 1.$$

Therefore,

$$32c^6 - 48c^4 + 18c^2 - 1 = 0.$$

Let

$$x = c^2.$$

Then

$$32x^3 - 48x^2 + 18x - 1 = 0.$$

We now factor the cubic. Observe that

$$32x^3 - 48x^2 + 18x - 1 = (2x - 1)(16x^2 - 16x + 1).$$

Hence,

$$(2x - 1)(16x^2 - 16x + 1) = 0.$$

Therefore either

$$2x - 1 = 0,$$

giving

$$x = \frac{1}{2},$$

or

$$16x^2 - 16x + 1 = 0.$$

Applying the quadratic formula,

$$x = \frac{16 \pm \sqrt{256 - 64}}{32} = \frac{16 \pm \sqrt{192}}{32}.$$

Since

$$\sqrt{192} = 8\sqrt{3},$$

we obtain

$$x = \frac{16 \pm 8\sqrt{3}}{32} = \frac{2 \pm \sqrt{3}}{4}.$$

Thus the three possible values of c^2 are

$$\frac{1}{2}, \quad \frac{2 + \sqrt{3}}{4}, \quad \frac{2 - \sqrt{3}}{4}.$$

Since $A = 15^\circ$ is acute and

$$\cos 15^\circ > \cos 30^\circ > \cos 45^\circ,$$

the appropriate value is the largest positive possibility:

$$\cos^2 15^\circ = \frac{2 + \sqrt{3}}{4}.$$

Hence,

$$\cos 15^\circ = \sqrt{\frac{2 + \sqrt{3}}{4}} = \frac{\sqrt{2 + \sqrt{3}}}{2}.$$

To put this into the standard radical form, observe that

$$(\sqrt{6} + \sqrt{2})^2 = 6 + 2 + 2\sqrt{12} = 8 + 4\sqrt{3} = 4(2 + \sqrt{3}).$$

Therefore,

$$\sqrt{2 + \sqrt{3}} = \frac{\sqrt{6} + \sqrt{2}}{2}.$$

Consequently,

$$\boxed{\cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}}.$$

We now use

$$\sin^2 A + \cos^2 A = 1.$$

Therefore,

$$\sin^2 15^\circ = 1 - \cos^2 15^\circ = 1 - \frac{2 + \sqrt{3}}{4}.$$

Hence,

$$\sin^2 15^\circ = \frac{2 - \sqrt{3}}{4}.$$

Since $\sin 15^\circ > 0$,

$$\sin 15^\circ = \sqrt{\frac{2 - \sqrt{3}}{4}} = \frac{\sqrt{2 - \sqrt{3}}}{2}.$$

To express this in standard radical form, observe that

$$(\sqrt{6} - \sqrt{2})^2 = 6 + 2 - 2\sqrt{12} = 8 - 4\sqrt{3} = 4(2 - \sqrt{3}).$$

Therefore,

$$\sqrt{2 - \sqrt{3}} = \frac{\sqrt{6} - \sqrt{2}}{2}.$$

Consequently,

$$\boxed{\sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}}.$$

Therefore, the required closed forms are

$$\boxed{\sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}}$$

and

$$\boxed{\cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}}.$$

The End